

7	(a)	Nuclear fission is the <u>splitting / disintegration of a heavy nucleus into two lighter nuclei with approximately equal masses.</u>		A1
	(b)	(i)	3 neutrons	A1
	(b)	(ii)	Mass difference including neutrons $= (234.993467 - 89.886883 - 142.905749 - 2 \times 1.008665) \times u$ $= 0.183505 \text{ u}$ [M1] Energy released $= 0.183505 \text{ u } c^2 = 2.742 \times 10^{-11} \text{ J}$ $= 2.742 \times 10^{-11} / (1.6 \times 10^{-19}) = 171.375$ [M1] $= 171 \text{ MeV}$ [A0]	
	(c)	(i)	Energy required from fission per second $= 500 \times 10^6 / 0.33 = 1.515 \times 10^9 \text{ J}$	C1



		Number of fissions per second required = $1.515 \times 10^9 / (2.742 \times 10^{-11})$ = 5.53×10^{19}	A1
	(c) (ii)	Mass of U-235 undergoing fission per second = $5.53 \times 10^{19} \times 234.99 \text{ u} = 2.156 \times 10^{-5} \text{ kg}$ [C1] Mass of U-235 that fissions per day = $2.156 \times 10^{-5} \times 24 \times 3600 = 1.863 = 1.9 \text{ kg}$ [A1]	
	(e)	Any two of the following: 1. <u>Only a small amount of uranium is required to produce a large amount of electrical energy.</u> This could remove Singapore's reliance on importing large amounts of fossil fuels. 2. <u>The use of nuclear energy is carbon neutral or less polluting unlike the use of fossil fuels.</u> Therefore this compares well with other carbon-neutral alternative energy sources. 3. <u>Nuclear power can potentially generate large amounts of energy to meet Singapore's needs.</u> 4. <u>Nuclear power requires less land area than other alternative energy sources such as solar or wind power.</u> This is suitable for land-scarce Singapore. 5. <u>Nuclear power is more reliable (can be generated 24 hours a day)</u> than other alternative energy sources such as wind or solar which depend on weather conditions. or other reasonable explanations.	B2

